

Computing Classical DH Parameters from a URDF

1 Overview

This document describes how to compute the classical (Spong) Denavit–Hartenberg parameters $(a_i, \alpha_i, d_i, \theta_i)$ for each joint i of a serial manipulator, starting from a URDF file.

Each joint’s DH row describes the homogeneous transform from frame $i-1$ to frame i :

$$T_i^{i-1} = R_z(\theta_i) T_z(d_i) T_x(a_i) R_x(\alpha_i).$$

The four parameters are:

- a_i : link length — distance between consecutive joint axes along the common normal.
- α_i : link twist — signed angle between consecutive joint axes, measured about $\hat{x}_i$.
- d_i : joint offset — signed distance along $\hat{z}_{i-1}$ from the previous frame origin to frame i ’s origin.
- θ_i : joint angle offset — signed rotation about $\hat{z}_{i-1}$ that aligns $\hat{x}_{i-1}$ with $\hat{x}_i$. For revolute joints, the runtime angle is $\theta_i + q_i$.

Notation convention. To avoid a common source of confusion, we use two distinct symbols throughout this document:

- $\hat{\mathbf{a}}_i$ denotes the direction of joint i ’s axis as extracted from the URDF (in the base frame).
- $\hat{\mathbf{z}}_i$ denotes the z -axis of DH frame i .

In classical DH, frame i has its z -axis along *joint* $i+1$ ’s axis, so $\hat{\mathbf{z}}_i = \hat{\mathbf{a}}_{i+1}$. To compute joint i ’s DH row, we need the geometry between axes i and $i+1$ — this is the single most common source of derivation errors.

2 Step 1: Extract Axis Data from the URDF

To compute the DH row for joint i , we need four values, all expressed in the base frame: $\mathbf{p}_i$, $\hat{\mathbf{a}}_i$, $\mathbf{p}_{i+1}$, $\hat{\mathbf{a}}_{i+1}$.

As we walk the kinematic chain, we maintain two running quantities:

- R_{cum} — cumulative rotation from base to the current link’s frame.
- $\mathbf{p}_{\text{cum}}$ — cumulative position of the current link’s origin, in the base frame.

Initial values (at the base): $R_{\text{cum}} = I$, $\mathbf{p}_{\text{cum}} = (0, 0, 0)$.

Suppose the URDF gives us, for each joint i :

- $\text{rpy} = (r_i, \beta_i, \gamma_i)$ — roll, pitch, yaw angles (URDF uses fixed-axis / extrinsic convention: $R = R_z(\gamma) R_y(\beta) R_x(r)$).
- $\text{xyz} = \mathbf{t}_i$ — translation of the joint's origin in the parent link's frame.
- $\text{axis} = \hat{\mathbf{a}}_i^{\text{local}}$ — unit direction of the joint's rotation axis, expressed in the child link's frame.

Computing $\mathbf{p}_1$ and $\hat{\mathbf{a}}_1$. Process joint 1 in three substeps, in order:

(a) Update the position: rotate the joint's translation by the *current* R_{cum} (i.e., the parent's rotation), then add:

$$\mathbf{p}_1 = \mathbf{p}_{\text{cum}} + R_{\text{cum}} \cdot \mathbf{t}_1 = (0, 0, 0) + I \cdot \mathbf{t}_1 = \mathbf{t}_1.$$

(b) Update the cumulative rotation:

$$R_{\text{cum}} \leftarrow R_{\text{cum}} \cdot R(r_1, \beta_1, \gamma_1) = I \cdot R(r_1, \beta_1, \gamma_1) = R(r_1, \beta_1, \gamma_1).$$

(c) Compute the joint axis direction, using the *updated* R_{cum} :

$$\hat{\mathbf{a}}_1 = R_{\text{cum}} \cdot \hat{\mathbf{a}}_1^{\text{local}}.$$

Computing $\mathbf{p}_2$ and $\hat{\mathbf{a}}_2$. After processing joint 1, the running state is:

$$R_{\text{cum}} = R(r_1, \beta_1, \gamma_1), \quad \mathbf{p}_{\text{cum}} = \mathbf{t}_1.$$

Apply the same three substeps to joint 2.

(a) Update the position: rotate joint 2's translation by the *current* R_{cum} (i.e., joint 1's rotation, since joint 1's child link is the parent of joint 2), then add:

$$\mathbf{p}_2 = \mathbf{p}_{\text{cum}} + R_{\text{cum}} \cdot \mathbf{t}_2 = \mathbf{t}_1 + R(r_1, \beta_1, \gamma_1) \cdot \mathbf{t}_2.$$

Unlike joint 1, this rotation is not guaranteed to be the identity, so the components of $\mathbf{t}_2$ may mix into $\mathbf{p}_2$.

(b) Update the cumulative rotation:

$$R_{\text{cum}} \leftarrow R_{\text{cum}} \cdot R(r_2, \beta_2, \gamma_2) = R(r_1, \beta_1, \gamma_1) \cdot R(r_2, \beta_2, \gamma_2).$$

(c) Compute the joint axis direction, using the *updated* R_{cum} :

$$\hat{\mathbf{a}}_2 = R_{\text{cum}} \cdot \hat{\mathbf{a}}_2^{\text{local}}.$$

Iterating. Continue this three-substep process for each subsequent joint along the kinematic chain. The output of Step 1 is a list of $(\mathbf{p}_i, \hat{\mathbf{a}}_i)$ pairs, one per joint, all in the base frame.

The key rule. At each joint, the translation $\mathbf{t}_i$ is rotated by the *parent's* cumulative rotation (because $\mathbf{t}_i$ is expressed in the parent's frame), while the axis $\hat{\mathbf{a}}_i^{\text{local}}$ is rotated by the *child's* cumulative rotation (because the axis is expressed in the child's frame, which includes the joint's own rpy).

3 Step 2: Choose Frame 0

Before computing any DH row, we establish frame 0 — the base frame of the DH chain. Frame 0 has two constraints and one free choice:

- $\hat{z}_0 = \hat{a}_1$ (required — frame 0's z -axis must lie along joint 1's axis).
- $\mathbf{o}_0$ must lie on joint 1's axis. Standard choice: $\mathbf{o}_0 = (0, 0, 0)$ if the URDF's base is already at the origin.
- $\hat{x}_0$ is free, as long as it is perpendicular to $\hat{z}_0$. Standard choice: project world $+x$ onto the plane perpendicular to $\hat{z}_0$, then normalize. If $\hat{z}_0 = (0, 0, 1)$, this gives $\hat{x}_0 = (1, 0, 0)$.

Complete the right-handed triad with $\hat{y}_0 = \hat{z}_0 \times \hat{x}_0$.

4 Step 3: Build Frame i from the Common Normal

Frame i is constructed from the common normal between axis i and axis $i+1$. We describe the process for joint 1 below; the procedure is identical for each subsequent joint with indices shifted.

Step 3a: Classify the axis pair. Compute

$$\mathbf{c} = \hat{\mathbf{a}}_1 \times \hat{\mathbf{a}}_2, \quad \mathbf{d}_{\text{pp}} = \mathbf{p}_2 - \mathbf{p}_1.$$

- If $\|\mathbf{c}\| > \epsilon$: axes are skew or intersecting (Case A / Case B).
- If $\|\mathbf{c}\| \leq \epsilon$: axes are parallel (Case C).

Use $\epsilon \approx 10^{-6}$ in practice.

Step 3b: Find the feet of the common normal. Cases A and B (non-parallel). We require the connecting vector $\mathbf{f}_2 - \mathbf{f}_1$ to be perpendicular to both axes. Writing $\mathbf{f}_1 = \mathbf{p}_1 + s \hat{\mathbf{a}}_1$ and $\mathbf{f}_2 = \mathbf{p}_2 + t \hat{\mathbf{a}}_2$, the two perpendicularity conditions

$$(\mathbf{f}_2 - \mathbf{f}_1) \cdot \hat{\mathbf{a}}_1 = 0, \quad (\mathbf{f}_2 - \mathbf{f}_1) \cdot \hat{\mathbf{a}}_2 = 0$$

give the 2×2 system:

$$\begin{bmatrix} 1 & -\hat{\mathbf{a}}_1 \cdot \hat{\mathbf{a}}_2 \\ \hat{\mathbf{a}}_1 \cdot \hat{\mathbf{a}}_2 & -1 \end{bmatrix} \begin{bmatrix} s \\ t \end{bmatrix} = \begin{bmatrix} \mathbf{d}_{\text{pp}} \cdot \hat{\mathbf{a}}_1 \\ \mathbf{d}_{\text{pp}} \cdot \hat{\mathbf{a}}_2 \end{bmatrix}.$$

Solve for s, t , then

$$\mathbf{f}_1 = \mathbf{p}_1 + s \hat{\mathbf{a}}_1, \quad \mathbf{f}_2 = \mathbf{p}_2 + t \hat{\mathbf{a}}_2.$$

Case C (parallel). The common normal is non-unique. By convention, project $\mathbf{p}_2$ onto axis 1:

$$\mathbf{f}_1 = \mathbf{p}_1 + (\mathbf{d}_{\text{pp}} \cdot \hat{\mathbf{a}}_1) \hat{\mathbf{a}}_1, \quad \mathbf{f}_2 = \mathbf{p}_2.$$

Step 3c: Determine $\hat{\mathbf{x}}_1$. Case A (skew, $\|\mathbf{f}_2 - \mathbf{f}_1\| > \epsilon$):

$$\hat{\mathbf{x}}_1 = \frac{\mathbf{f}_2 - \mathbf{f}_1}{\|\mathbf{f}_2 - \mathbf{f}_1\|}.$$

Case B (intersecting, feet coincide):

$$\hat{\mathbf{x}}_1 = \frac{\hat{\mathbf{a}}_1 \times \hat{\mathbf{a}}_2}{\|\hat{\mathbf{a}}_1 \times \hat{\mathbf{a}}_2\|}.$$

There is still a sign ambiguity; choose the sign so $\hat{\mathbf{x}}_1 \cdot \hat{\mathbf{x}}_0 \geq 0$ (this tends to keep θ_1 near zero).

Case C (parallel):

$$\hat{\mathbf{x}}_1 = \frac{\mathbf{p}_2 - \mathbf{f}_1}{\|\mathbf{p}_2 - \mathbf{f}_1\|}.$$

If $\mathbf{p}_2 = \mathbf{f}_1$ (the axes are coincident, not merely parallel), this formula is degenerate. Fall back to $\hat{\mathbf{x}}_0$ projected into the plane perpendicular to $\hat{\mathbf{a}}_2$, then normalized.

Step 3d: Assemble frame 1.

$$\mathbf{o}_1 = \mathbf{f}_1, \quad \hat{\mathbf{z}}_1 = \hat{\mathbf{a}}_2, \quad \hat{\mathbf{x}}_1 = \text{as computed above}, \quad \hat{\mathbf{y}}_1 = \hat{\mathbf{z}}_1 \times \hat{\mathbf{x}}_1.$$

From this point forward, $\hat{\mathbf{z}}_1$ refers to the DH frame axis (which equals $\hat{\mathbf{a}}_2$), not the joint 1 axis.

5 Step 4: Compute the DH Parameters for Joint 1

With frame 0 (Step 2) and frame 1 (Step 3) both fully defined, apply the four formulas:

$$a_1 = (\mathbf{f}_2 - \mathbf{f}_1) \cdot \hat{\mathbf{x}}_1 \tag{1}$$

$$\alpha_1 = \text{atan2}\left((\hat{\mathbf{z}}_0 \times \hat{\mathbf{z}}_1) \cdot \hat{\mathbf{x}}_1, \hat{\mathbf{z}}_0 \cdot \hat{\mathbf{z}}_1\right) \tag{2}$$

$$d_1 = (\mathbf{o}_1 - \mathbf{o}_0) \cdot \hat{\mathbf{z}}_0 \tag{3}$$

$$\theta_1 = \text{atan2}\left((\hat{\mathbf{x}}_0 \times \hat{\mathbf{x}}_1) \cdot \hat{\mathbf{z}}_0, \hat{\mathbf{x}}_0 \cdot \hat{\mathbf{x}}_1\right) \tag{4}$$

For a revolute joint, θ_1 is a constant offset; the runtime joint angle is $\theta_1 + q_1$, where q_1 is the commanded joint variable. For a prismatic joint, d_1 is the constant offset and the runtime distance is $d_1 + q_1$.

Validation

Before trusting the result, check:

- $a_1 \geq 0$ in Cases A and B (by construction, if $\hat{\mathbf{x}}_1$ was chosen as the foot-to-foot direction).
- Compose $T_1^0 T_2^1 \dots T_n^{n-1}$ at the zero configuration and confirm the resulting end-effector pose matches the URDF forward kinematics. Any mismatch signals a sign error somewhere in the chain.

6 Step 5: Iterate to the Next Joint

To compute joint 2's DH row, repeat Steps 3 and 4:

- Use frame 1 (just built) in place of frame 0.
- Build frame 2 from the common normal between axis 2 ($\hat{\mathbf{a}}_2$) and axis 3 ($\hat{\mathbf{a}}_3$).
- Apply the four DH formulas with the indices shifted by one.

Continue until all n joints have been processed.

The last joint. For joint n , there is no axis $n+1$. The z -axis of frame n is still constrained to lie along joint n 's axis ($\hat{\mathbf{z}}_n = \hat{\mathbf{a}}_n$), but the origin placement along that axis and the direction of $\hat{\mathbf{x}}_n$ are free. Standard practice is to place $\mathbf{o}_n$ at the tool center point and choose $\hat{\mathbf{x}}_n$ to match the tool frame's x -axis, both supplied by the URDF's `tool0` fixed joint.

7 Worked Example: Joint 1 of the Motoman GP12

This example illustrates **Case A (skew axes)**.

The URDF declares:

```
<joint name="joint_1_s" type="revolute">
  <origin rpy="0 0 0" xyz="0 0 0.450"/>
  <axis xyz="0 0 1"/>
</joint>
<joint name="joint_2_l" type="revolute">
  <origin rpy="0 0 0" xyz="0.155 0 0"/>
  <axis xyz="0 1 0"/>
</joint>
```

Step 1 (axis data). All rpy values are zero, so $R_{\text{cum}} = I$ throughout and axis directions pass through unchanged. Cumulative positions simplify to the sum of xyz translations:

$$\begin{aligned} \mathbf{p}_1 &= (0, 0, 0.450), & \hat{\mathbf{a}}_1 &= (0, 0, 1) \\ \mathbf{p}_2 &= (0.155, 0, 0.450), & \hat{\mathbf{a}}_2 &= (0, 1, 0) \end{aligned}$$

Step 2 (frame 0).

$$\mathbf{o}_0 = (0, 0, 0), \quad \hat{\mathbf{z}}_0 = (0, 0, 1), \quad \hat{\mathbf{x}}_0 = (1, 0, 0), \quad \hat{\mathbf{y}}_0 = (0, 1, 0).$$

Step 3a (classification). $\mathbf{c} = \hat{\mathbf{a}}_1 \times \hat{\mathbf{a}}_2 = (0, 0, 1) \times (0, 1, 0) = (-1, 0, 0)$, with $\|\mathbf{c}\| = 1 > \epsilon$. Case A (skew).

Step 3b (feet of the common normal). $\mathbf{d}_{\text{pp}} = (0.155, 0, 0)$, $\hat{\mathbf{a}}_1 \cdot \hat{\mathbf{a}}_2 = 0$, so the system becomes:

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} s \\ t \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies s = 0, t = 0.$$

$$\mathbf{f}_1 = (0, 0, 0.450), \quad \mathbf{f}_2 = (0.155, 0, 0.450).$$

Step 3c ($\hat{x}_1$). $\hat{x}_1 = (0.155, 0, 0)/0.155 = (1, 0, 0)$.

Step 3d (assemble frame 1).

$$\mathbf{o}_1 = (0, 0, 0.450), \quad \hat{\mathbf{z}}_1 = \hat{\mathbf{a}}_2 = (0, 1, 0), \quad \hat{\mathbf{x}}_1 = (1, 0, 0), \quad \hat{\mathbf{y}}_1 = (0, 0, -1).$$

Step 4 (DH parameters).

$$a_1 = (\mathbf{f}_2 - \mathbf{f}_1) \cdot \hat{\mathbf{x}}_1 = (0.155, 0, 0) \cdot (1, 0, 0) = 0.155$$

$$\begin{aligned} \alpha_1 &= \text{atan2}((\hat{\mathbf{z}}_0 \times \hat{\mathbf{z}}_1) \cdot \hat{\mathbf{x}}_1, \hat{\mathbf{z}}_0 \cdot \hat{\mathbf{z}}_1) \\ &= \text{atan2}((-1, 0, 0) \cdot (1, 0, 0), 0) = \text{atan2}(-1, 0) = -\frac{\pi}{2} \end{aligned}$$

$$d_1 = (\mathbf{o}_1 - \mathbf{o}_0) \cdot \hat{\mathbf{z}}_0 = (0, 0, 0.450) \cdot (0, 0, 1) = 0.450$$

$$\theta_1 = \text{atan2}((\hat{\mathbf{x}}_0 \times \hat{\mathbf{x}}_1) \cdot \hat{\mathbf{z}}_0, \hat{\mathbf{x}}_0 \cdot \hat{\mathbf{x}}_1) = \text{atan2}(0, 1) = 0$$

Result for joint 1:

$$a_1 = 0.155 \text{ m}, \quad \alpha_1 = -\frac{\pi}{2}, \quad d_1 = 0.450 \text{ m}, \quad \theta_1 = 0.$$

8 Worked Example: Joint 2 of the Motoman GP12

This example illustrates **Case C (parallel axes)** and reuses frame 1 as computed in the previous example.

The URDF declares:

```
<joint name="joint_2_l" type="revolute">
  <origin rpy="0 0 0" xyz="0.155 0 0"/>
  <axis xyz="0 1 0"/>
</joint>
<joint name="joint_3_u" type="revolute">
  <origin rpy="0 0 0" xyz="0 0 0.614"/>
  <axis xyz="0 -1 0"/>
</joint>
```

Step 1 (axis data). Continuing the cumulative sum:

$$\begin{aligned} \mathbf{p}_2 &= (0.155, 0, 0.450), & \hat{\mathbf{a}}_2 &= (0, 1, 0) \\ \mathbf{p}_3 &= (0.155, 0, 1.064), & \hat{\mathbf{a}}_3 &= (0, -1, 0) \end{aligned}$$

Note that $\hat{\mathbf{a}}_3 = -\hat{\mathbf{a}}_2$. The URDF deliberately assigns opposite signs to the shoulder and elbow axis directions — physically this doesn't change anything (an axis line is symmetric under sign flip), but it does affect the DH table. A positive $\hat{\mathbf{a}}_3$ would produce $\alpha_2 = 0$ instead of $\alpha_2 = \pi$; the final forward kinematics would still match.

Frame 1 (from the previous example).

$$\mathbf{o}_1 = (0, 0, 0.450), \quad \hat{\mathbf{z}}_1 = (0, 1, 0), \quad \hat{\mathbf{x}}_1 = (1, 0, 0), \quad \hat{\mathbf{y}}_1 = (0, 0, -1).$$

Step 3a (classification).

$$\mathbf{c} = \hat{\mathbf{a}}_2 \times \hat{\mathbf{a}}_3 = (0, 1, 0) \times (0, -1, 0) = (0, 0, 0), \quad \|\mathbf{c}\| = 0.$$

Case C (parallel). The axes are antiparallel but that is still “parallel” for classification: lines in 3D have no inherent direction, so the common normal is non-unique regardless of the sign of $\hat{\mathbf{a}}_3$.

Step 3b (feet of the common normal, parallel convention). Set $\mathbf{f}_3 = \mathbf{p}_3$ and project $\mathbf{p}_3$ onto axis 2:

$$\begin{aligned} \mathbf{d}_{\text{pp}} &= \mathbf{p}_3 - \mathbf{p}_2 = (0, 0, 0.614), & \mathbf{d}_{\text{pp}} \cdot \hat{\mathbf{a}}_2 &= 0. \\ \mathbf{f}_2 &= \mathbf{p}_2 + 0 \cdot \hat{\mathbf{a}}_2 = (0.155, 0, 0.450), & \mathbf{f}_3 &= (0.155, 0, 1.064). \end{aligned}$$

Step 3c ($\hat{\mathbf{x}}_2$, parallel convention).

$$\hat{\mathbf{x}}_2 = \frac{\mathbf{p}_3 - \mathbf{f}_2}{\|\mathbf{p}_3 - \mathbf{f}_2\|} = \frac{(0, 0, 0.614)}{0.614} = (0, 0, 1).$$

Step 3d (assemble frame 2).

$$\mathbf{o}_2 = (0.155, 0, 0.450), \quad \hat{\mathbf{z}}_2 = \hat{\mathbf{a}}_3 = (0, -1, 0), \quad \hat{\mathbf{x}}_2 = (0, 0, 1), \quad \hat{\mathbf{y}}_2 = \hat{\mathbf{z}}_2 \times \hat{\mathbf{x}}_2 = (-1, 0, 0).$$

Step 4 (DH parameters).

$$\begin{aligned} a_2 &= (\mathbf{f}_3 - \mathbf{f}_2) \cdot \hat{\mathbf{x}}_2 = (0, 0, 0.614) \cdot (0, 0, 1) = 0.614 \\ \alpha_2 &= \text{atan2}((\hat{\mathbf{z}}_1 \times \hat{\mathbf{z}}_2) \cdot \hat{\mathbf{x}}_2, \hat{\mathbf{z}}_1 \cdot \hat{\mathbf{z}}_2) \\ &= \text{atan2}(\mathbf{0} \cdot (0, 0, 1), -1) = \text{atan2}(0, -1) = \pi \\ d_2 &= (\mathbf{o}_2 - \mathbf{o}_1) \cdot \hat{\mathbf{z}}_1 = (0.155, 0, 0) \cdot (0, 1, 0) = 0 \\ \theta_2 &= \text{atan2}((\hat{\mathbf{x}}_1 \times \hat{\mathbf{x}}_2) \cdot \hat{\mathbf{z}}_1, \hat{\mathbf{x}}_1 \cdot \hat{\mathbf{x}}_2) \\ &= \text{atan2}((0, -1, 0) \cdot (0, 1, 0), 0) = \text{atan2}(-1, 0) = -\frac{\pi}{2} \end{aligned}$$

Result for joint 2:

$$a_2 = 0.614 \text{ m}, \quad \alpha_2 = \pi, \quad d_2 = 0, \quad \theta_2 = -\frac{\pi}{2}.$$

Interpretation of $\theta_2 = -\pi/2$. The URDF’s zero configuration places joint 2 “straight up” (link 2 oriented along world $+z$), but the DH frame convention puts $\hat{\mathbf{x}}_2$ along the common normal — which also points along world $+z$. The $-\pi/2$ offset rotates $\hat{\mathbf{x}}_1$ (world $+x$) to $\hat{\mathbf{x}}_2$ (world $+z$) about $\hat{\mathbf{z}}_1$ (world $+y$). At runtime, the commanded angle q_2 is added to this offset: $\theta_2^{\text{runtime}} = -\pi/2 + q_2$.

This offset is *not* a bug — it is an artifact of the URDF and DH conventions disagreeing about what the zero pose looks like. Published DH tables for the GP12 typically bake this offset into the joint angle definition, so the “canonical” and “URDF” zero poses differ by exactly this quarter turn.

9 Sign-Convention Pitfalls

The most common errors in DH derivations come from sign ambiguities. Three rules prevent most of them:

1. **The four DH parameters are not independent scalars — they are the components of a rigid transform.** Flipping the sign of $\hat{x}_i$ flips both a_i and α_i , and shifts θ_i by π . You cannot flip one in isolation.
2. **Commit to a convention for $\hat{x}_i$ before computing.** The “common normal points from axis i to axis $i+1$ ” convention (used above) gives $a_i \geq 0$ and tends to produce clean tables.
3. **Validate by composing the transforms.** At the zero configuration, the product $T_1^0 \cdots T_n^{n-1}$ must match the end-effector pose computed directly from the URDF. If it doesn’t, there is a sign error, and it is almost always in α_i or θ_i .